SUMMER INTERNSHIP REPORT

On

ReliefRoute: Smart Disaster Resource Allocation Simulator

Submitted in

Department of Information Technology

B.Tech Fourth Year (2026-2027)

Submitted by

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Under the guidance of

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(Recognized by AICTE, Govt. of India & Affiliated to Dr. A.P.J. Abdul Kalam Technical University, Lucknow)

AKTU College Code - 054

CERTIFICATE

This is to certify that **Tulika Srivastava** (University Roll Number: 2300540130055), a student of B.Tech Fourth Year, Department of Information Technology, Babu Banaras Das Institute of Technology & Management, Lucknow, has successfully completed the Research Internship from 2 June 2026 to 15 August 2026 at the Indian Institute of Science (IISc), Bengaluru.

During this period, the student worked on the project **"Analysis of Photonic Devices Using Analytical, Simulation and Neural Network Approaches"** under the guidance of Prof. T. Srinivas. This report presents the mini project **"ReliefRoute: Smart Disaster Resource Allocation Simulator"** and has been prepared as part of the requirement for the B.Tech degree. It is the student's own work.

Faculty Guide
Mr. Arvind Gautam

Head of Department
Dr. Santosh Kumar Shukla, IT

Date: 02/11/26

Place: Lucknow

TRAINING OBJECTIVE

The main aim of this internship was to learn how photonic devices can be studied using three complementary approaches: analytical models, numerical simulation and neural networks. The internship was carried out at the Indian Institute of Science (IISc), Bengaluru, under the guidance of Prof. T. Srinivas, and the project was titled "Analysis of Photonic Devices Using Analytical, Simulation and Neural Network Approaches".

Alongside the internship, a Python mini project named ReliefRoute was developed to apply the same habits of modelling, testing and visualising results to a practical problem: sharing limited relief supplies fairly among several disaster-affected areas. When a disaster happens, food, water and medical kits are always limited, and deciding quickly who gets how much is difficult. ReliefRoute is a simulator that supports this decision using a transparent, rule-based allocation method.

The objectives were:

1. To understand the working of basic photonic devices and the equations that describe them.
2. To build analytical models and compare them with simulation results.
3. To learn how a neural network can be trained to predict the response of a device from its parameters.
4. To write clean Python code for data handling, modelling and plotting.
5. To design a decision-support tool (ReliefRoute) that allocates food, water and medical kits across affected areas according to urgency and population.
6. To build an interactive web application with Streamlit, with editable inputs, tables, charts and a downloadable CSV report.
7. To document the work in a proper technical report.

ABOUT TRAINING ORGANIZATION

The internship was done at the Indian Institute of Science (IISc), Bengaluru. The completion certificate is dated 10 September 2026 and is signed by Prof. T. Srinivas.

The Indian Institute of Science is one of India's leading institutions for advanced scientific and technological research and higher education. It was established in 1909 in Bengaluru through the vision of Jamsetji Nusserwanji Tata, with the support of the Maharaja of Mysore and the Government of India. Over more than a century it has grown into a research university with many departments and centres covering science, engineering and interdisciplinary areas, and it has been recognised as an Institution of Eminence.

IISc offers research internships in which students work on live research problems under faculty guidance. This gives undergraduate students the chance to learn research methods, to use simulation and computing tools, and to see how theory is turned into results.

Vision

To be a world-class institution for research and higher education that produces knowledge and trained people for the benefit of the nation and society.

Mission

- To carry out high-quality research in science, engineering and allied fields.
- To train students and researchers in rigorous, independent thinking.
- To encourage collaboration between disciplines and with industry.
- To give young students exposure to real research through internships and projects.

- **Name of the organization:** Indian Institute of Science (IISc), Bengaluru
- **Address:** C. V. Raman Avenue, Bengaluru, Karnataka 560012, India
- **Website:** www.iisc.ac.in
- **Internship project:** Analysis of Photonic Devices Using Analytical, Simulation and Neural Network Approaches
- **Certificate signed by:** T. Srinivas, Professor
- **Duration of internship:** 2 June 2026 to 15 August 2026

Indian Institute of Science, Bangalore

10 Sept 2026

Completion Certificate

This is to certify that **Tulika Srivastava**, a student of B.Tech in Information Technology at BBDITM, Lucknow, has successfully completed her internship at the **Indian Institute of Science (IISc), Bengaluru**, from **2nd June 2026 to 15th August 2026**. She worked on the project titled "**Analysis of Photonic Devices Using Analytical, Simulation and Neural Network Approaches.**" During the internship, she showed keen interest and dedication towards the work and successfully completed her project with good performance and sincere effort.

T. Srinivas, Professor

Figure 1: Internship Completion Certificate

LEARNING OUTCOMES

During the research internship at IISc, I gained the following skills:

- Understanding of the basic working of photonic devices and how their behaviour depends on design parameters.
- Use of analytical models to estimate device behaviour quickly.
- Use of numerical simulation and comparison of simulated results with analytical results.
- Training and testing a neural network to predict device response from input parameters.
- Python programming for data generation, analysis and plotting.
- Reading research material, writing technical documentation and presenting results.

Through the ReliefRoute project I also learned the following things:

- **Python programming:** I wrote the full application in Python, organising the code into small functions for sample data, allocation and rounding, and a Streamlit interface on top.
- **Algorithm design:** I learned how to turn a real-life question ("who should get how much?") into a clear rule. The priority weight is urgency multiplied by one plus the normalised affected population.
- **Constrained allocation:** I learned weighted water-filling, where no area can receive more than it asked for and leftover stock is shared again among areas that still have unmet demand.
- **Rounding without losing totals:** I used the largest-remainder method so that whole-number allocations still add up to the available stock.
- **Data handling with Pandas and NumPy:** I cleaned user input, converted text to numbers, clipped urgency to the range 1 to 5 and calculated shortages and coverage.
- **Building web apps with Streamlit:** I built a sidebar for stock inputs, an editable data table, metric cards, an expandable explanation box and a download button, and used session state to keep the data between clicks.
- **Data visualization:** I used Plotly Express to draw grouped bar charts for allocated resources and unmet demand.
- **Responsible use of AI-style tools:** I learned to state the limits of a simulator clearly. The app says plainly that it uses simulated data and is not meant for real emergency operations.

DETAILED SCHEDULE OF TRAINING

Week / Date	Activity
Week 1	Orientation, discussion of the project with the guide and a literature survey on photonic devices
Weeks 2 - 3	Study of the theory and derivation of the analytical models for the chosen devices
Week 4	Setting up the Python environment and coding the analytical models
Weeks 5 - 6	Setting up numerical simulations and running parameter sweeps
Week 7	Comparing analytical and simulation results and checking the differences
Weeks 8 - 9	Preparing the dataset and designing the neural network model
Week 10	Training, testing and tuning the neural network and comparing all three approaches
Week 11	Final analysis, documentation and presentation of results to the guide
Along with internship	Designing the ReliefRoute mini project: problem study, allocation rule and sample scenario
Along with internship	Coding the allocation functions, Streamlit interface, charts and CSV export
Along with internship	Testing with different stock levels and urgency values, writing the README and preparing this report

DESCRIPTION ABOUT PROJECT WORK

1. Project Title

ReliefRoute: Smart Disaster Resource Allocation Simulator

Type: Academic prototype using simulated data

2. Introduction

ReliefRoute is a web-based decision-support simulator built with Streamlit. It shows how a limited stock of food packets, drinking water and medical kits can be shared among several disaster-affected areas. The user enters the available stock and edits a table of areas, with each area's number of affected people, urgency level from 1 to 5 and estimated needs. The application then calculates how much each area should receive, how much demand is still unmet, and how much of the total demand is covered. Two strategies are available, a priority-weighted one that favours urgent and larger areas, and a proportional fair-share one that follows each area's requested quantity.

3. Features

- Editable table of affected areas with rows that can be added or removed
- Sidebar inputs for available food, water and medical kits
- Two allocation strategies: priority-weighted and proportional fair-share
- Urgency scale from 1 (lower) to 5 (critical)
- No area is given more than its estimated need
- Whole-number allocations that add up exactly to the available stock
- Shortage (unmet demand) shown for every area and every resource
- Summary metrics: food, water, medical kits supplied and overall demand coverage
- Interactive Plotly charts for allocated resources and unmet demand
- Download of the allocation report as a CSV file
- Built-in explanation of the allocation logic and a clear disclaimer

4. Technologies Used

Part	Technology
Frontend	Streamlit (page layout, sidebar, data editor, metric cards, expander, download button)
Backend	Python 3.10 or newer
Data processing	Pandas and NumPy
Visualization	Plotly Express (grouped bar charts)
Libraries	streamlit >= 1.35, pandas >= 2.0, numpy >= 1.24, plotly >= 5.18
Data	Simulated sample scenario, editable by the user

5. Project Structure

The project is small and has three files:

- **app.py** contains the whole application: the page design, the sample data function sample_data(), the allocation function allocate_one(), the rounding function round_allocations(), the results dashboard and the CSV export.
- **requirements.txt** lists the libraries that must be installed.
- **README.md** explains how to run the project, the demo workflow, the algorithm and the important limitation.

6. System Architecture

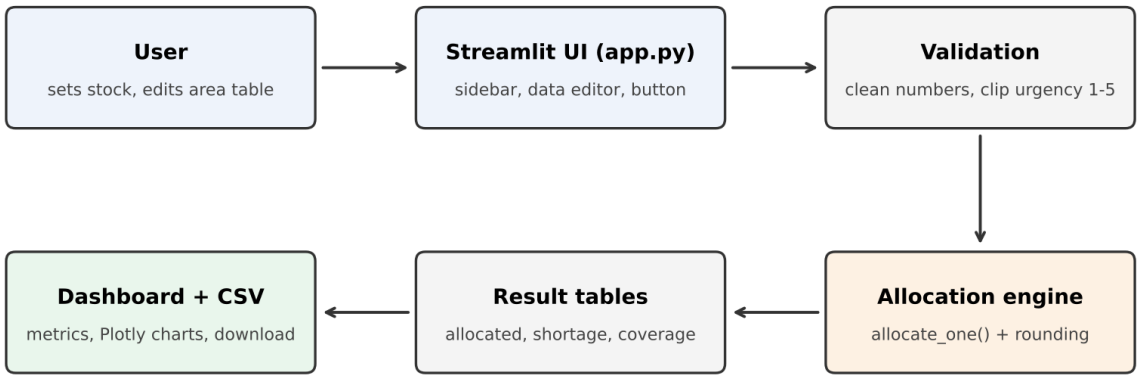


Figure 2: System architecture

The user sets the stock and edits the area table in the Streamlit interface. The inputs are cleaned and validated, the allocation engine calculates the quantities for food, water and medical kits, and the results are shown as tables, metric cards and charts, with a CSV download.

7. Working of the Project

The system works in the following steps:

1. **Set the stock.** In the sidebar the user enters the available food packets, water in litres and medical kits.
2. **Choose the strategy.** The user selects Priority-weighted greedy or Proportional fair-share.
3. **Edit the areas.** The table lists each area with people affected, urgency (1 to 5) and the food, water and medical kit needs. The starter scenario has four areas: Riverside, Old Town, Central Ward and Hillview.
4. **Validate the data.** On clicking Generate Allocation Plan, the app checks that the table is not empty and every area has a name. Numbers are converted safely, missing values become 0 and urgency is limited to 1 to 5.
5. **Calculate priority weights.** For the priority strategy, each area gets a weight equal to urgency x (1 + people affected / highest people affected). For fair-share, the weight is simply the area's need.
6. **Distribute the stock.** The stock is divided according to the weights. If an area would receive more than its need, it is capped and the remaining stock is shared again among the other areas. This repeats until the stock is finished or all needs are met.
7. **Round the values.** The decimal results are converted to whole numbers using the largest-remainder method so that the total still equals the available stock.
8. **Build the result table.** The app adds allocated quantity and shortage for food, water and medical kits.
9. **Show the dashboard.** Metric cards, the result table and two bar charts are displayed.
10. **Download.** The final table can be downloaded as a CSV file.

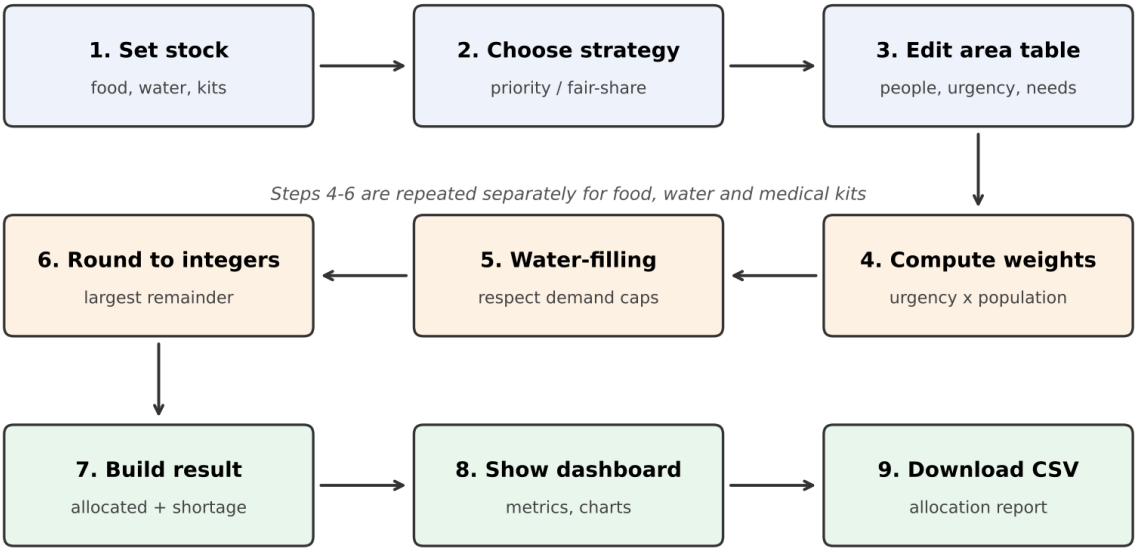


Figure 3: Flowchart of the working

The same allocation routine is called three times, once each for food, water and medical kits, so every resource is distributed using the same rule.

8. Input Data (Sample Scenario)

The starter scenario used for demonstration has four areas and the following stock: 800 food packets, 1,800 litres of water and 220 medical kits.

Area	People affected	Urgency (1-5)	Food need (packets)	Water need (litres)	Medical kits need
Riverside	800	5	500	1200	100
Old Town	450	4	300	700	80
Central Ward	650	3	400	900	90
Hillview	300	5	200	500	60
Total demand	2,200	-	1,400	3,300	330

The total demand (1,400 packets, 3,300 litres and 330 kits) is much higher than the available stock, so the allocation rule decides who is served first.

9. Allocation Logic

Priority-weighted strategy. The priority weight of area i is:

$$weight(i) = urgency(i) \times (1 + people(i) / max\ people)$$

An area with higher urgency or more affected people gets a larger share. Areas with zero need get no share. Each area is capped at its own need, and any stock left over is redistributed among areas that are still short. The weights for the sample scenario, and the resulting share of the 800 food packets, are:

Area	Urgency	1 + people / max	Weight	Share of stock	Food packets
Riverside	5	2.0000	10.0000	35.0%	280
Old Town	4	1.5625	6.2500	21.9%	175
Central Ward	3	1.8125	5.4375	19.0%	152
Hillview	5	1.3750	6.8750	24.1%	193

The decimal results (280.09, 175.05, 152.30 and 192.56) are rounded to 280, 175, 152 and 193, which add up to exactly 800.

Proportional fair-share strategy. The weight of each area is just its requested quantity, so every area receives roughly the same percentage of what it asked for, again with demand caps.

Rounding. Each value is first rounded down. The leftover units are then given, one at a time, to the areas with the largest fractional parts, without exceeding any area's need.

10. Dashboard and Output

After clicking Generate Allocation Plan, the application shows:

- **Metric cards:** food supplied, water supplied, medical kits supplied and overall demand coverage.
- **Result table:** need, allocated quantity and shortage for every area and resource.
- **Allocated resources chart:** grouped bar chart of food, water and medical kits per area.
- **Remaining unmet demand chart:** grouped bar chart of the shortages.
- **Download button:** the full table as a CSV file (reliefroute_allocation_report.csv).
- **Explanation box:** an expandable section describing the allocation logic.

For the sample scenario, the four metric cards show:

Metric	Supplied / Needed	Coverage
Food supplied	800 / 1,400	57.1%
Water supplied	1,800 / 3,300 L	54.5%
Medical kits supplied	220 / 330	66.7%
Overall demand coverage	2,820 / 5,030 (all units combined)	56.1%

Result of the priority-weighted strategy (allocated / needed):

Area	Food (packets)	Water (litres)	Medical kits	Coverage
Riverside	280 / 500	630 / 1200	77 / 100	54.8%
Old Town	175 / 300	394 / 700	48 / 80	57.1%
Central Ward	152 / 400	343 / 900	42 / 90	38.6%
Hillview	193 / 200	433 / 500	53 / 60	89.3%

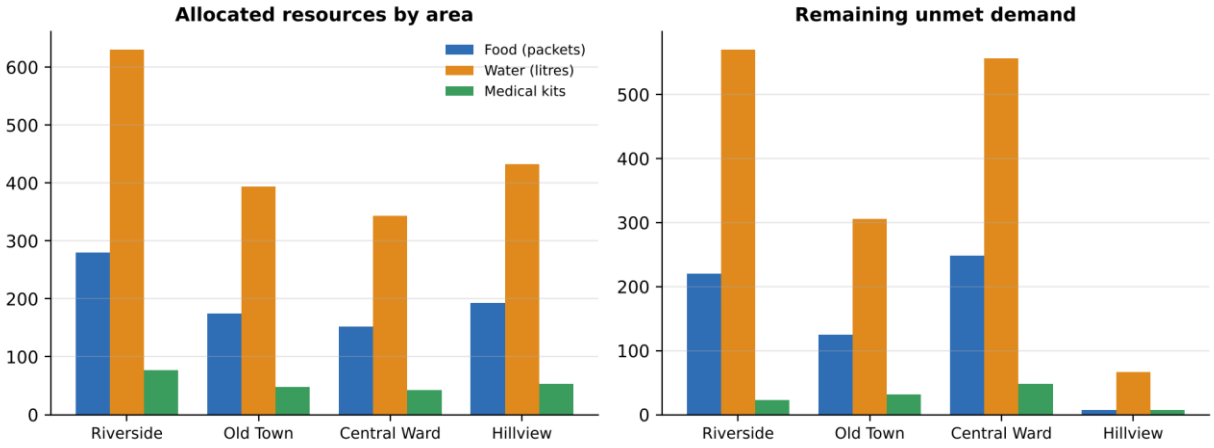


Figure 4: Allocated resources by area and remaining unmet demand (priority-weighted strategy)

Result of the proportional fair-share strategy (allocated / needed):

Area	Food (packets)	Water (litres)	Medical kits	Coverage
Riverside	286 / 500	654 / 1200	67 / 100	55.9%
Old Town	171 / 300	382 / 700	53 / 80	56.1%
Central Ward	229 / 400	491 / 900	60 / 90	56.1%
Hillview	114 / 200	273 / 500	40 / 60	56.2%

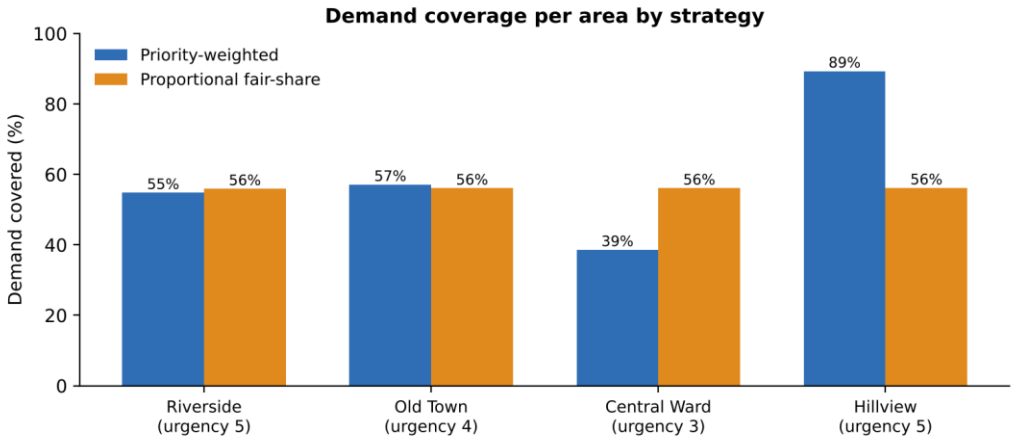
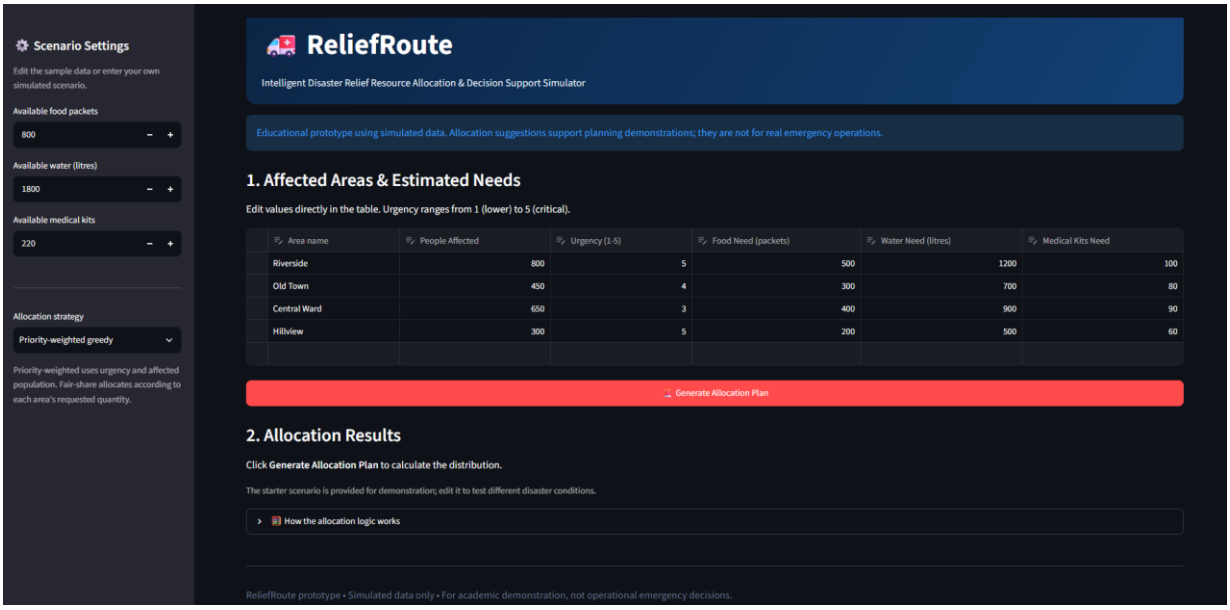


Figure 5: Demand coverage per area for the two strategies

Observations. Both strategies distribute the same total stock, so the overall coverage is the same (56.1%). The difference is in who benefits. Fair-share gives every area about 56% of its demand. The priority-weighted strategy shifts supplies towards the small but critical area Hillview (urgency 5), which receives about 89% of its demand, while Central Ward (urgency 3) receives about 39%. This shows how the priority rule protects the most urgent areas when stock is short. The figures in this section were produced by running the same allocation functions as the application on the default scenario.





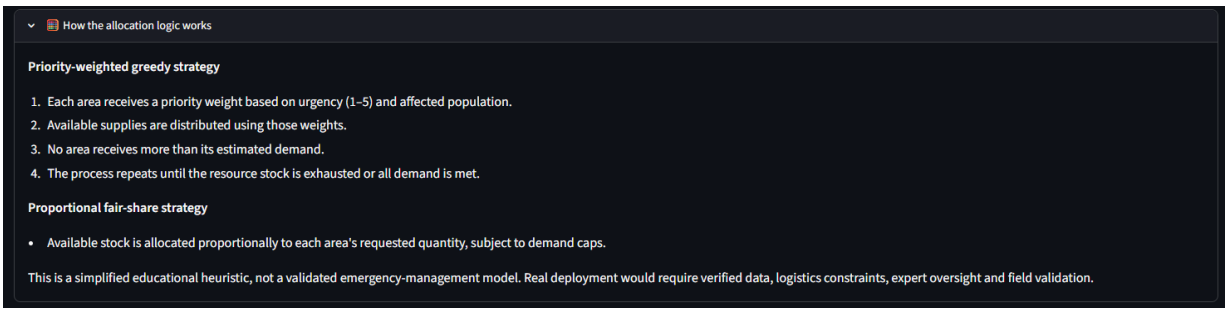


Figure 6: Dashboard

11. Installation and Running

1. Install Python 3.10 or newer.
2. Open a terminal in the project folder and, if wanted, create and activate a virtual environment.
3. Install the libraries: pip install -r requirements.txt
4. Run the app: streamlit run app.py
5. Open the local address shown in the terminal, edit the scenario and click Generate Allocation Plan.

12. Limitations and Future Improvements

ReliefRoute is a simplified educational heuristic. It uses simulated data and has not been validated for real emergency management. Real use would need verified data, transport and road constraints, expert supervision and field testing.

Possible improvements:

- Road and distance data to plan delivery routes and vehicle loads
- Live map view of the affected areas
- More resources and priority groups such as children, elderly and hospitals
- Saving and comparing several scenarios
- Import of area data from CSV or Excel files
- User login and role-based access
- PDF report generation

SCOPE OF PROJECT WITH SDG MAPPING

Scope of the Project

- **For disaster management teams and NGOs:** It can be used in training and planning exercises to see how different allocation rules change who receives help.

- **For colleges and students:** It is a simple example of how Python, Pandas and Streamlit can be used to build a decision-support tool.

- **For local administrations:** With real data and validation, a similar tool could help prepare relief stock plans before a flood, earthquake or cyclone.

- **For researchers:** The two strategies can be compared, and new rules can be added and tested easily.

- **Future scope:** Adding routing, maps, more resource types and report generation can turn the prototype into a fuller planning assistant.

SDG Mapping



SDG	How the project relates to it
SDG 11: Sustainable Cities and Communities	Supports planning for disaster response and reduces the impact of disasters on affected communities.
SDG 13: Climate Action	Helps prepare for floods, storms and other climate-related hazards by improving readiness and response planning.
SDG 2: Zero Hunger	Plans the sharing of food packets so that the most urgent areas are served first.
SDG 3: Good Health and Well-being	Includes medical kits as a separate resource, so health needs are considered along with food and water.
SDG 6: Clean Water and Sanitation	Includes drinking water as a resource and shows where water needs are still unmet.
SDG 9: Industry, Innovation and Infrastructure	Uses modern open-source software tools to build a digital decision-support solution.

CONCLUSION

The research internship at the Indian Institute of Science gave me a good introduction to studying photonic devices with analytical, simulation and neural network approaches, and showed me how research is planned, tested and documented.

The ReliefRoute project shows how a clear rule-based method can support the sharing of limited relief supplies. The application takes the stock and the needs of each area, ranks areas by urgency and affected population, avoids over-allocation, keeps the totals correct after rounding and shows the unmet demand openly. The results for the sample scenario show that the priority-weighted strategy protects critical areas like Hillview, while the fair-share strategy treats all areas equally in percentage terms.

Working on this project helped me learn Python, Pandas, NumPy, Plotly and Streamlit, and also how to explain an algorithm and state its limits. The tool is only a planning aid with simulated data, and real emergency decisions must always be taken by trained people using verified information. In the future, features such as route planning, maps and report generation can make it more useful.

REFERENCES

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2. Pandas Documentation: https://pandas.pydata.org/docs/
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